

Bike Rental Data Analysis

Project Objective

- Analysis bike rental patterns .
- Understand user behavior .
- Identify trends (time , weather) .

Dataset Overview

- Total dataset - 3 .
- Final dataset – merged data .
- Dataset 3 used for dashboard due to higher granularity .

Data Cleaning

- Removed duplicates .
- Handles missing values .
- Fixed data format .
- Rename columns (casual , registered) .

Data Transformation

- Created Month name column.
- Converted numeric values to categories.
- Created Weekend / Weekday column.
- Structured data for analysis.

Dashboard Overview

Month_name  

Jan

Feb

Weekend/Week...  

Weekday

Weekend

Hour_d3  

0

1

2

3

4

5

6

7

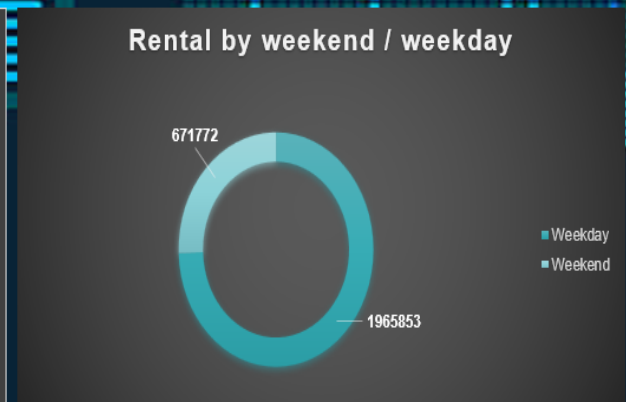
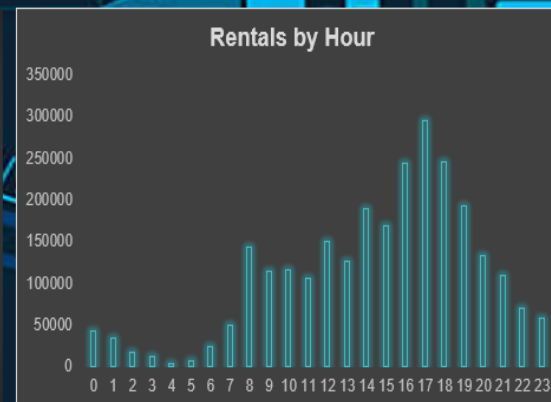
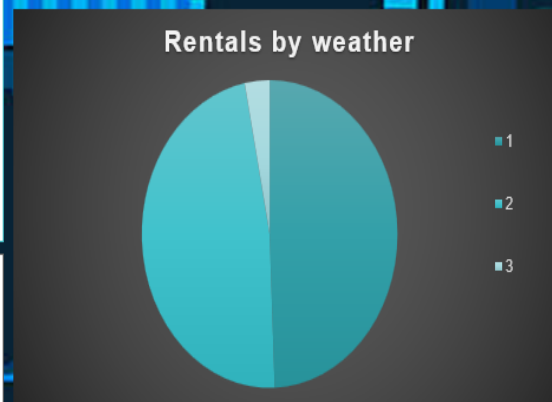
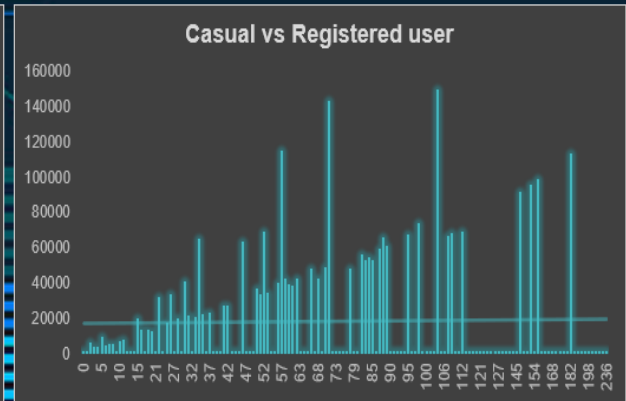
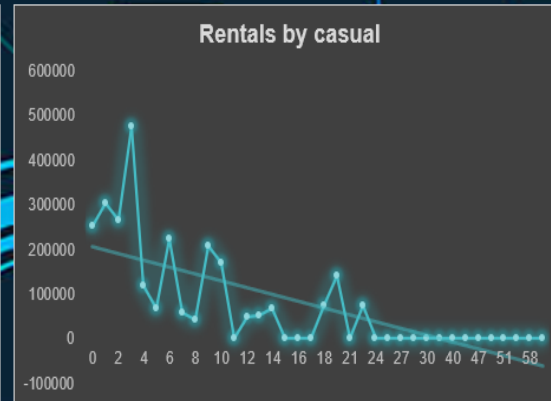
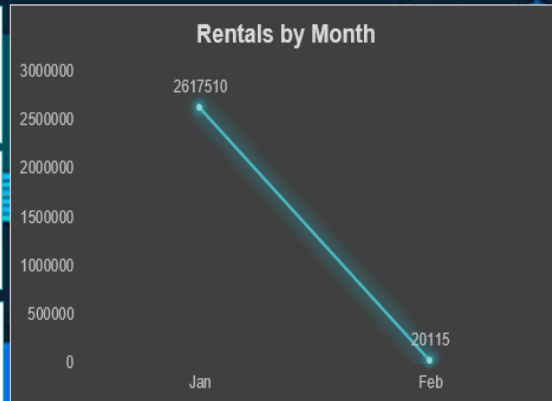
8

weathersit_d3  

1

2

3



- Interactive dashboard .
- Filters: Month, Weather .
- Multiple visual insights .

Key Visuals

- **Rentals by Month.**
- **Rentals by Hours.**
- **Rentals by weather.**
- **User Type analysis.**

Key Insights

- Rentals peak during evening hour .
- Weekday usage higher than weekend .
- Clear weather = highest rentals .
- Registered users dominated .

Conclusion

- Demand depends on time and weather .
- Business can optimize bike availability.
- Data helps in decision making .